

Report on CPM and PERT

Name: Rajat Prakash Dhal

Email: 1hk22cs115@hkbk.edu.in

USN: 1hk22cs115

Introduction

CPM (Critical Path Method) and PERT (Program Evaluation and Review Technique) are important project scheduling techniques used to plan, organize, and control complex projects. Both methods help in identifying task sequences, estimating project duration, and managing dependencies effectively.

Objective of the Report (CPM and PERT)

The objective of this report is to study and understand the principles, structure, and application of **Critical Path Method (CPM)** and **Program Evaluation and Review Technique (PERT)** as project scheduling techniques.

This report aims to:

- Explain the concepts and methodology of CPM and PERT.
- Understand how project networks are constructed and analyzed.
- Identify the critical path and calculate project duration.
- Compare deterministic (CPM) and probabilistic (PERT) approaches.
- Evaluate their importance in effective project planning, scheduling, and control.

Overall, the report seeks to develop a clear understanding of how CPM and PERT help in managing time, resources, and risks in project management.

Critical Path Method (CPM)

Overview

CPM is a deterministic project scheduling technique where task durations are known and fixed. It identifies the **longest sequence of dependent tasks**, known as the *critical path*, which determines the minimum project completion time.

Key Concepts

- Activities and dependencies
- Forward pass (earliest start/finish time)
- Backward pass (latest start/finish time)
- Float (slack time)
- Critical path (zero float)

Features

- Suitable for predictable projects
- Focuses on time-cost optimization
- Helps identify tasks that cannot be delayed

Applications

- Construction projects
- Manufacturing processes
- Engineering projects

Advantages

- Clear project timeline
- Identifies critical activities
- Helps in resource allocation

Limitations

- Assumes fixed task durations
- Not suitable for highly uncertain projects

Program Evaluation and Review Technique (PERT)

Overview

PERT is a probabilistic scheduling technique used when task durations are uncertain. It uses three time estimates to calculate expected completion time.

Three Time Estimates

- Optimistic time (O)
- Most likely time (M)
- Pessimistic time (P)

Expected Time Formula:

$$TE = \frac{O + 4M + P}{6}$$

Features

- Handles uncertainty
- Uses probability-based estimates
- Focuses on research and development projects

Applications

- R&D projects
- Software development
- Innovative or experimental projects

Advantages

- Accounts for uncertainty
- Improves risk assessment
- Provides realistic time estimation

Limitations

- More complex calculations
- Requires reliable estimates

Comparison Between CPM and PERT

Basis	CPM	PERT
Nature	Deterministic	Probabilistic
Time Estimates	Single estimate	Three estimates (O, M, P)
Focus	Time & cost control	Time & risk analysis
Best For	Predictable projects	Uncertain projects

Conclusion

CPM and PERT are both effective project scheduling techniques used to manage complex projects.

- **CPM** is best suited for projects with fixed timelines and predictable activities.
- **PERT** is more suitable for projects involving uncertainty and risk.

Together, they provide structured planning, better time management, and improved project control in various organizational environments.